

- 1 City A is in Sweden and is located at (58°N , 16°E). Sydney, in Australia, is located at (33°S , 151°E).

Robert lives in Sydney and needs to give an online presentation to his colleagues in City A starting at 5:00 pm Thursday, local time in Sweden.

What time and day, in Sydney, should Robert start his presentation?

It is given that $15^{\circ} = 1$ hour time difference.

Ignore daylight saving. (3 marks)

(Q20, 2021 HSC) Medium

- 2 The Coordinated Universal Time (UTC) of Auckland is +12 hours and the UTC of Chicago is -5 hours.

When the time in Chicago is 2 pm, Thursday, what is the time in Auckland?

- A 9 pm, Wednesday B 7 am, Thursday
C 9 pm, Thursday D 7 am, Friday (1 mark)

(Q5, 2019 HSC) Medium

- 3 A plane leaves Miami, Florida (UTC-4) at 2:40 pm on Monday and arrives in Copenhagen, Denmark (UTC+2) at 6:55 am on Tuesday. If the flight distance is 7840 km what is the average speed of the aeroplane, to the nearest km/h? (2 marks)

Bonus question (see page iv) Hard

- 4 Holly flies from Auckland (UTC+12) across the International Date Line to Honolulu (UTC-10). The plane departs Auckland at 9:20 pm Friday and flies 7155 kilometres at an average speed of 810 km/h.

What is the local time when the plane lands in Honolulu? (3 marks)

Bonus question Hard

- 5 A plane left Dubai (UTC+4) at 1:30 pm on Thursday. It flew non-stop to Dallas (UTC-5) at an average speed of 810 km/h, arriving in Dallas at 8:45 pm on Thursday. What was the distance the plane flew? (2 marks)

Bonus question Hard

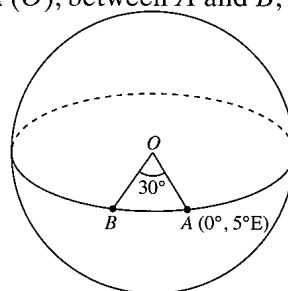
- 6 The time in Brisbane is $4\frac{1}{2}$ hours ahead of the time in New Delhi. John flew from New Delhi to Brisbane via Singapore. His plane left New Delhi at 11.30 am (New Delhi time),

stopped for 3 hours in Singapore, and arrived in Brisbane at 9.00 am the following day (Brisbane time).

What was the plane's total flying time? (3 marks)

(Q29a, 2018 HSC) Medium

- 7 Island A and island B are both on the equator. Island B is west of island A. The longitude of island A is 5°E and the angle at the centre of Earth (O), between A and B, is 30° .



NOT TO SCALE

- i What is the longitude of island B? (1 mark) Hard
ii What time is it on island B when it is 10 am on island A? (1 mark) Medium

(Q27d, 2017 HSC)

- 8 Melbourne is located at (38°S , 145°E) and Dubai is located at (24°N , 55°E).

- i Calculate the difference in longitude between Melbourne and Dubai. (1 mark) Easy
ii Show that the time difference between Melbourne and Dubai is 6 hours. (1 mark) Easy
iii A plane leaves Melbourne on Friday at 11.30 pm. The flight time to Dubai is 15 hours. What will be the time and the day in Dubai when the plane is due to land? (2 marks) Easy

(Q27e, 2016 HSC)

- 9 Stockholm is located at 59°N 18°E and Darwin is located at 13°S 131°E .

What is the time difference between Stockholm and Darwin? (Ignore time zones and daylight saving.)

- A 184 minutes B 288 minutes
C 452 minutes D 596 minutes (1 mark)

(Q14, 2015 HSC) Easy

- 10 Singapore is located at 1°N 104°E and Sydney is located at 34°S 151°E .

HSC Practice Questions

Working with Time

What is the time difference between Singapore and Sydney? (Ignore daylight saving.) (2 marks)

(Q26g, 2014 HSC) **Medium**

- 11 Karin is in Athens, which is two hours ahead of Greenwich Mean Time. Marco is in New York, which is five hours behind Greenwich Mean Time.
- Karin is going to ring Marco at 10 pm on Tuesday, Athens time. What day and time will it be in New York when she rings? (1 mark) **Medium**
 - Marco is going to fly from New York to Athens. His flight will leave on Wednesday at 9 am, New York time, and will take 11 hours. What day and time will it be in Athens when he arrives? (2 marks) **Easy**

(Q27e, 2013 HSC)

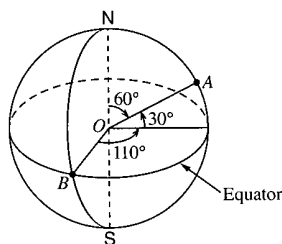
- 12 Perth in Western Australia is 8 hours ahead of Greenwich in England. Cape Town in South Africa is 2 hours ahead of Greenwich.

What is the time in Cape Town when it is 1 pm in Perth?

- A 3 am B 7 am
C 7 pm D 11 pm (1 mark)

(Q3, 2011 HSC) **Easy**

- 13 In this diagram of the Earth, O represents the centre and B lies on both the Equator and the Greenwich Meridian.



NOT TO SCALE

What is the latitude and longitude of point A ?

- A $30^\circ\text{N } 110^\circ\text{E}$ B $30^\circ\text{N } 110^\circ\text{W}$
C $60^\circ\text{N } 110^\circ\text{E}$ D $60^\circ\text{N } 110^\circ\text{W}$ (1 mark)

(Q15, 2010 HSC) **Easy**

- 14 Lee wants to call home to Sydney (34°S , 151°E) from Papeete, Tahiti (17°S , 149°W) when it is 7 pm on Friday in Sydney.

Give the time and day in Papeete when she should make the call. (Ignore time zones.) (2 marks) **Medium**

(Q25c ii, 2010 HSC)

- 15 Osaka is at 34°N , 135°E , and Denver is at 40°N , 105°W .

- Show that there is a 16-hour time difference between the two cities. (Ignore time zones.) (2 marks) **Medium**
- John lives in Denver and wants to ring a friend in Osaka. In Denver it is 9 pm Monday. What time and day is it in Osaka then? (1 mark) **Easy**
- John's friend in Osaka sent him a text message which happened to take 14 hours to reach him. It was sent at 10 am Thursday, Osaka time. What was the time and day in Denver when John received the text? (2 marks) **Hard**

(Q26b, 2009 HSC)

- 16 Kim lives in Perth (32°S , 115°E). He wants to watch an ice hockey game being played in Toronto (44°N , 80°W) starting at 10.00 pm on Wednesday.

What is the time in Perth when the game starts?

- A 9.00 am on Wednesday
B 7.40 pm on Wednesday
C 12.20 am on Thursday
D 11.00 am on Thursday (1 mark)

(Q20, 2007 HSC) **Medium**

- 17 Cassie flew from London (52°N , 0°E) to Manila (15°N , 120°E).

Her plane left London at 9.30 am Monday (London time), stopped for 5 hours in Singapore and arrived in Manila at 4.00 pm Tuesday (Manila time).

What was the total flying time? (Ignore time zones.) (3 marks)

(Q26d, 2006 HSC) **Medium**

- 18 The location of Town A is $25^\circ\text{N } 45^\circ\text{E}$. The location of Town B is $10^\circ\text{N } 105^\circ\text{E}$.

Which of the following is true? (Ignore time zones.)

- A Town A is four hours behind Town B .
B Town A is four hours ahead of Town B .
C Town A is one hour behind Town B .
D Town A is one hour ahead of Town B .

(1 mark)

(Q19, 2005 HSC) **Easy**

- 19 Kathmandu is 30° west of Perth. Using the longitude difference, what is the time in Kathmandu when it is noon in Perth?

- A 10:00 am B 11:30 am
C 12:30 pm D 2:00 pm (1 mark)

(Q10, 2003 HSC) **Easy**

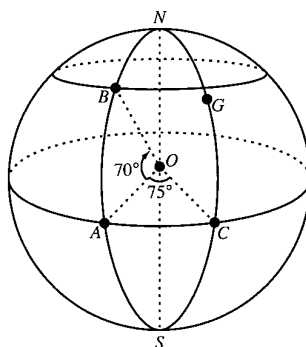
- 20 The table shows the approximate coordinates for two cities.

City	Latitude	Longitude
Buenos Aires	35° S	60° W
Adelaide	35° S	140° E

- What is the time difference between Adelaide and Buenos Aires? (Ignore time zones.) (1 mark) **Medium**
- Roy lives in Adelaide and his cousin Juan lives in Buenos Aires. Roy wants to telephone Juan at 7 pm on a Friday night, Buenos Aires time. At what time, and on what day, should Roy make the call? (2 marks) **Medium**

(Q25b, 2002 HSC)

- 21 In this diagram of the Earth, O represents the centre and G represents Greenwich. The point A lies on the equator.



NOT TO SCALE

- What is the time difference between Greenwich and point A ? (Ignore time zones.) (1 mark) **Medium**
- What is the latitude of point B ? (1 mark) **Easy**

(Q24b, 2001 HSC)

- 1 Difference in longitude
 $= 151^\circ - 16^\circ$
 $= 135^\circ$ ✓
 Time difference $= (135 \div 15)$ h
 $= 9$ h ✓
 Now Sydney is 9 hours ahead of A.
 So the time in Sydney is 9 h after 5 pm Thursday.
 The time is 2:00 am Friday. ✓
 (3 marks)

- 2 5 hours after 2 pm is 7 pm.
 12 hours after 7 pm is 7 am the next day.
 The time in Auckland is 7 am Friday.
 Answer D

- 3 $4 + 2 = 6$
 Copenhagen is 6 hours ahead of Miami.
 Plane lands at 6:55 am minus 6 h
 $= 12:55$ am Miami time.
 Time of flight
 $= 2:40$ pm Mon. to 12:55 am Tues.
 $= 10$ h 15 min
 $= 10.25$ h ✓
 $\text{Speed} = \frac{7840}{10.25}$
 $= 764.8780488...$
 $= 765$ (nearest whole) ✓
 $\therefore$ the average speed is 765 km/h.
 (2 marks)

- 4 Time of flight $= 7155 \div 810$
 $= 8\frac{5}{6}$ h
 $= 8$ h 50 min ✓
 Honolulu is 22 hours behind Auckland.
 9:20 pm plus 8 h 50 min is
 6:10 am Saturday (Auckland time). ✓
 Subtracting 22 hours gives
 8:10 am Friday. ✓ (3 marks)

- 5 $4 + 5 = 9$
 $\therefore$ time difference = Dallas is 9 hours behind Dubai.
 Dubai 1:30 pm Thurs. = Dallas 4:30 am Thurs.
 Flight time = 4:30 am to 8:45 pm
 $= 16$ h 15 min
 $= 16.25$ h ✓
 Distance = Speed $\times$ Time
 $= 810 \times 16.25$
 $= 13162.5$
 $\therefore$ the distance was 13162.5 km. ✓ (2 marks)

- 6 $4\frac{1}{2}$ h ahead of 11:30 am is 4 pm. ✓

So the plane left New Delhi at 4 pm Brisbane time.
 From 4.00 pm until 9.00 am the following day is 17 hours. ✓
 Flying time $= (17 - 3)$ h
 $= 14$ h ✓
 (3 marks)

- 7 i B is 30° west of A.
 The longitude of A is 5° E, so 5° west of A is the prime meridian.
 Now $30 - 5 = 25$ so B is 25° west of the prime meridian.
 So the longitude of B is 25° W. (1 mark)
 ii Angular difference $= 30^\circ$
 Time difference
 $= (30 \div 15)$ h
 $= 2$ h
 B is west of A so it is 2 h behind A.
 The time on island B would be 8 am. (1 mark)

- 8 i Difference in longitude
 $= 145^\circ - 55^\circ$
 $= 90^\circ$ (1 mark)
 ii Every 15° difference in longitude is 1 hour difference in time.
 So time difference
 $= (90 \div 15)$ h
 $= 6$ h (1 mark)
 iii The plane leaves at 11:30 pm Melbourne time.
 Dubai is 6 h behind Melbourne time so the plane leaves at 5:30 pm Dubai time. ✓
 It lands 15 hours later.
 15 hours after 5:30 pm is 8:30 am the next day.
 So the plane is due to land at 8:30 am Saturday Dubai time. ✓ (2 marks)

- 9 Difference in longitude
 $= 131^\circ - 18^\circ$
 $= 113^\circ$
 Time difference
 $= 113 \times 4$ minutes
 $= 452$ minutes ✓
 Answer C (1 mark)

- 10 Singapore (1° N, 104° E),
 Sydney (34° S, 151° E)
 Angular difference in longitude
 $= 151^\circ - 104^\circ$
 $= 47^\circ$ ✓

Time difference $= 47 \times 4$ min
 $= 188$ min
 $= 3$ h 8 min
 (Ignoring time zones) ✓
 (2 marks)

- 11 i New York is 7 hours behind Athens time.
 So 10 pm Athens time is 3 pm New York time.
 In New York it will be 3 pm on Tuesday when Karin rings. (1 mark)
 ii At 9 am Wednesday in New York it will be 4 pm Wednesday in Athens. ✓
 The flight takes 11 hours.
 So 11 hours after 4 pm will be 3 am.
 Marco will arrive in Athens at 3 am on Thursday. ✓
 (2 marks)

- 12 When it is 1 pm in Perth it is 5 am in Greenwich. So it will be 7 am in Cape Town.
 Answer B

- 13 A is 30° north of the Equator and 110° east of the Greenwich meridian.
 The coordinates of A are 30° N 110° E.
 Answer A

- 14 Difference in longitude
 $= 151^\circ + 149^\circ$
 $= 300^\circ$
 Time difference
 $= (300 \div 15)$ hours
 $= 20$ hours ✓
 Papeete is 20 hours behind Sydney time.
 Lee should call home when it is 11 pm Thursday in Papeete. ✓
 (2 marks)

- 15 i Longitude difference
 $= 135^\circ + 105^\circ$
 $= 240^\circ$ ✓
 Time difference
 $= 240^\circ \div 15$
 $= 16$ hours ✓
 (2 marks)

- ii Denver is 16 hours behind Osaka as it is further west.
 $\therefore$ 9 pm Monday in Denver is 9 pm Monday + 16 hours in Osaka, i.e. 1 pm Tuesday in Osaka.

(1 mark)

- iii Osaka is 16 hours ahead of Denver. ✓
 $\therefore$ John received the text, 10 am Thursday – 16 hours + 14 hours ✓
 i.e. 8 am Thursday in Denver.

(2 marks)

- 16 Longitude difference
 $= 115^\circ + 80^\circ$
 $= 195^\circ$

Time difference
 $= 195^\circ \div 15^\circ$
 $= 13$ hours

$\therefore$ Perth is 13 hours ahead of Toronto.

$\therefore$ The time will be 11:00 am on Thursday.

Answer D

- 17 London (52°N , 0°E)
 Manila (15°N , 120°E)

Longitude difference $= 120^\circ$
 $\therefore$ Time difference $= 120 \div 15$
 $= 8$ hours ✓

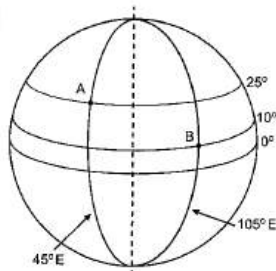
Now 4 pm Tuesday in Manila is 8 am Tuesday in London.

$\therefore$ Total time taken is $22\frac{1}{2}$ hours (from Monday 9:30 am to Tuesday 8 am London time), including 5 hours stopping time. ✓

$\therefore$ Total flying time
 $= 22\frac{1}{2} - 5$
 $= 17\frac{1}{2}$ hours ✓

(3 marks)

18



Since $15^\circ = 1$ hour,
 time difference
 $=$ longitude difference $\div 15$
 $= (105 - 45) \div 15$
 $= 60 \div 15$
 $= 4$ hours

B is ahead 4 hours as it is further east.

Answer A

- 19 Time difference $= 30^\circ \times 4$ min
 $= 120$ min
 $= 2$ h

$\therefore$ Kathmandu is 2 h behind Perth, i.e. 10:00 am.

Answer A

- 20 i Longitude difference
 $= 60^\circ + 140^\circ$
 $= 200^\circ$

Time difference
 $= 200 \div 15$
 $= 13.\bar{3}$ h
 $= 13$ h 20 min

(1 mark)

- ii Adelaide is 13 h 20 min ahead of Buenos Aires as it is to the east. ✓

$\therefore$ 7 pm Fri, Buenos Aires time is (7 pm Fri + 13 h 20 min) Adelaide time, i.e. 8:20 am Saturday, Adelaide time. ✓

(2 marks)

- 21 i Time difference $= 75^\circ \div 15$
 $= 5$ hours
 (1 mark)

- ii 70°N

(1 mark)